

Preprocessing and Feature Engineering Report

Beyond Accuracy: A Clinical Validation Approach to Cardiovascular Disease Prediction Using Machine Learning

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DSE6311 Capstone

This project evaluates machine learning model performance through a clinical validation lens rather than standard accuracy metrics. To assess clinically appropriate risk, how should model performance be evaluated, given the trade-off between sensitivity and specificity?

Model optimization and validation will balance the delicate trade-off between false negatives and false positives. The final model may not be the most accurate based solely on accuracy/ROC-AUC metrics.

A tuned ensemble method, such as Random Forest or XGBoost optimized for sensitivity, will outperform Logistic Regression on recall and ROC-AUC. The optimal model will be defined as the sensitivity-specificity combination that best aligns with clinical risk thresholds and patient outcomes.

Methods:

The original plan for preprocessing prior to formal EDA was to handle missing values in ca and thal features, scale and standardize continuous variables, and encode categorical variables. A train/test split will then be performed, and SMOTE will be explored for class imbalance. PCA would be assessed for dimensionality reduction.

After performing EDA, several findings caused a shift in the original preprocessing plan. Class imbalance was found to be milder than expected, at a ratio of 54% no disease and 46% with disease. Cholesterol has a significant outlier at 564mg/dl, and oldpeak shows enough skewness to consider transformation. Missing values were confirmed in 6 rows, ca and thal features, and a decision will be made to drop rather than impute after further assessment. Correlation analysis showed a mild multicollinearity, and PCA may not be necessary, but it will still be assessed as part of the project. Thalach and oldpeak emerged as the strongest continuous predictors, and ca, exang, thal, and cp emerged as the strongest categorical predictors, which will influence feature importance. Since no severe multicollinearity was found, the original feature set will move forward mostly intact.

The dataset was split into training and test sets prior to any preprocessing steps to prevent data leakage. The split ratio was calculated using a mathematically derived formula based on the number of predictors (Credit: Katherine Geist, PhD, DSE6311). This resulted in a 72/28 training/test split, which gave 218 training records and 85 test records. Stratified splitting was applied and confirmed to maintain the same class distribution as the original dataset.

Categorical variables were assessed for encoding requirements prior to modeling. Binary variables (sex, fbs, and exang) were already encoded as 0/1 and did not require further transformation. Ordinal variables, ca and slope, kept their numeric coding as the values carry a meaningful clinical order. One-hot encoding was applied to cp, restecg, and thal using pandas get_dummies. Their numeric codes do not hold meaningful weight but do represent clinical categories. Encoding was applied to training and test sets separately to prevent data leakage, and

column alignment was used to confirm consistency across both sets.

To assess for missingness, a formal Little's MCAR test was implemented via the `nanianar` package in R using the `rp2` bridge in Google Colab. Code attributions were noted: [njtierney/nanianar](https://github.com/njtierney/nanianar): [Tidy data structures, summaries, and visualisations for missing data](https://github.com/njtierney/nanianar/blob/master/README.md).

The null hypothesis states that if the data are MCAR (Missing Completely At Random), the p-value is greater than 0.05. If the p-value is below 0.05, then the null hypothesis is not supported, and the data may not be MCAR, and other considerations will need to be assessed.

After performing the MCAR test on the full UCI dataset, the p-value of 0.8986 indicates that the null hypothesis is supported, and the missing values are Missing Completely At Random. The rows with missingness in columns `ca` and `thal` can be dropped from both training and test sets without introducing bias. Imputation would be considered clinically inappropriate due to the significance of the values in `ca` (vessel blockage), and `thal` (thalassemia type).

After MCAR confirmation, the 6 total rows with missingness were dropped from the training and test sets. After removing missing rows, the dataset contains 297 records, with 217 in the training set and 80 in the test set.

While the impact on the training set was very low 0.5% data loss, there is a large impact (5.9%) on the testing set as 5 of the 6 rows (83%) were in that dataset. Total overall data loss was at a low of 2%, but there is a disproportionate loss among the testing data. This is noted as a limitation given the already small original test dataset size (85 → 80 records).

It was noticed that missingness removal should have been done prior to splitting the data, and would have provided a more even distribution of data between the test and training sets.

Therefore, to correct this error, the rows were dropped from the original UCI data, and a clean dataset was created, re-split, and re-encoded using the cleaned version. (code was left in place for reference but commented out)

After splitting a cleaned dataset, the result was 297 records, split into 213 for training and 84 for testing. Both sets maintained consistent class distributions of 54% no CVD and 46% CVD present, confirming that the stratified split worked correctly. This distribution will provide a better result and will have recovered 4 records into the test set.

Considering logistic regression will be used for the baseline model, Shapiro-Wilks tests were run to assess normality, and Q-Q plots were generated to visually confirm the findings. Box-Cox transformation was selected for `oldpeak` and `cholesterol` based on the strong right skew for `oldpeak` and the extreme outlier in `cholesterol`. A shift of +1 was applied to Box – Cox for `oldpeak` due to values of 0 in that column.

During preprocessing, two related challenges were identified with the variables for ST segmentation, `oldpeak` and `slope`. `Oldpeak` showed significant zero inflation and applying a Box – Cox transformation improved but did not resolve the issue. Prior EDA identified a moderate correlation between `oldpeak` and `slope` (0.58), suggesting that these variables captured overlapping information. Clinical literature review revealed that these statistical findings reflect a

known psychological relationship of ST depression and myocardial ischemia as established in the ACC/AHA Exercise Testing Guidelines (Gibbons et al 1997, 2002). This insight informed a feature engineering decision to combine the two variables into an ordinal severity index with four defined categories. This method resolves both the zero-inflation and multicollinearity issues, while producing a more clinically meaningful feature than two standalone variables.

Dimensionality reduction was achieved through clinically informed feature engineering rather than a pure mathematical method such as PCA. To address the multicollinearity identified in EDA and the zero-inflation problem in oldpeak, slope and oldpeak were combined into the variable st_severity. This approach was used to preserve clinical interpretability, which is a requirement for clinical decision support tools.

Supervised feature selection will be performed during the modeling phase. After initial model training, feature importance scores from Random Forest and XGBoost will be used to identify which features contribute most to the prediction. Features identified as low importance will be candidates for removal in subsequent model iterations.

Scaling will be performed on continuous variables, age, trestbps, chol_boxcox, and thalach. They will be fit and transformed on training data only, then applied to the test set without re-fitting. SMOTE will be explored for mild class imbalance and PCA for dimensionality reduction.

Results:

Shapiro-Wilk testing on continuous variables (age, thalach, trestbps, chol, oldpeak) showed all variables with a p-value under 0.05, indicating that each does not follow a normal distribution. However, statistics show that age, thalach, and trestbps hover around 0.98 suggesting that they are close to normal distribution. Cholesterol shows a statistic of 0.932, and oldpeak 0.836, confirming that both are not normally distributed. Shapiro-Wilk is known to be more sensitive with larger data sets. To further back up findings, Q-Q plots were created to visually assess Shapiro-Wilk testing (see Figure 7).

Age and maximum heart rate (age, thalach) followed the normal line closely and were considered practically normal in Shapiro – Wilk. Oldpeak showed the most pronounced departure from normality, with points clustering near zero, followed by a sharp rise, consistent with the skewness observed in EDA. Cholesterol showed a mostly normal fit, with an extreme point in the upper tail corresponding to the 564mg/dL outlier. Based on these findings, Box-Cox transformation was applied to oldpeak and cholesterol, while age, thalach, and trestbps proceeded without transformation.

After the Box – Cox transformation, cholesterol responded well and oldpeak had some partial improvement. Cholesterol statistic went from 0.932 $\rightarrow$ 0.994 and a p-value of 0.000 $\rightarrow$ 0.505, indicating a successful transformation. Oldpeak statistic went from 0.836 $\rightarrow$ 0.878, no change in

p-value, and remains non-normal due to zero-inflation. Zero inflation occurs when a large proportion of patients, in this case those with no ST depression, record a zero (see Figure 8).

As stated in the methods section, creation of an ST severity variable was used to address this issue, while simultaneously addressing the multicollinearity found between slope and oldpeak. The st_severity variable was then validated against the target variable to confirm clinical validity. CVD prevalence increased with increasing categorical severity from Category 0 (negative) at 31.1% to Category 3 (strongly positive) at 87.2%. Category 1 (equivocal) showed similar CVD prevalence to Category 0 at 28.3%, validating the clinical literature that upsloping ST depression is inconclusive. These findings confirm that the engineered feature captures a clinical signal and is expected to be a strong predictor during modeling.

*****Note on Limitations:** *The st_severity variable is specific to exercise stress test interpretation and captures exercise induced ST depression morphology per ACC/AHA guidelines. It does not capture ST elevation findings, which are a distinct clinical finding associated with acute myocardial infarction, which is not evaluated in this dataset****

StandardScaler was applied to continuous variables, age, trestbps, chol_boxcox, and thalach. Scaling was fit on the training set only and then applied to both the training and test sets to prevent data leakage. Descriptive statistics confirmed standardization with mean of 0 and standard deviations of 1 on all four variables.

SMOTE was applied to the training set as a proof-of-concept exploration of handling class imbalance. The minority class was oversampled from 98 to 115 records through synthetic patient generation and produced a balanced 50/50 split of 230 training records. Due to the mild class imbalance (54%/46%), the impact of SMOTE performance will be compared to the training set not impacted by SMOTE during the modeling phase.

PCA was applied to the four continuous variables after standardization. The analysis showed that the variance was relatively evenly distributed across the four principal components. The variables explained 38.6%, 27.7%, 20.4%, and 13.3%, respectively. All four components were required to reach 95% cumulative explained variance. A scree plot was created and no clear elbow point was visible for the dimensionality reduction cutoff (see Figure 9). This finding suggests that PCA does not offer meaningful dimensionality reduction for this feature set thus supporting the decision to retain the original engineered features for modeling. It should be noted that PCA was applied only to continuous variables, as the dataset contains a mix of continuous and categorical variables. A mixed method of unsupervised learning could be applied such as PCA for continuous and a cluster method for categorical, or Factor Analysis of Mixed Data (FAMD), could be more appropriate. Given the time and complexity of project deadlines and criteria, these methods were not explored.

Variance Inflation Factor (VIF) analysis confirmed that any multicollinearity was resolved across the final feature set. All VIF values fell below the 5.0 threshold, with the highest value being sex

at 4.01. The reduction of oldpeak and slope into the st_severity engineered feature showed a VIF of 2.65 resolving the two variable correlation from EDA.

Discussion:

Several meaningful findings were found during the preprocessing and feature engineering portion of this project that both confirmed and refined the original analysis plan. Little's MCAR Test confirmed that the 6 missing values from ca and thal were random and could be removed without introducing bias. Shapiro-Wilk testing and Q-Q plots identified zero inflation in oldpeak and confirmed skewness in cholesterol (see Figure 7 and 8). A Box-Cox transformation was applied to cholesterol and further investigation was done for oldpeak. The investigation on this variable provided the most significant finding of the week; that there was a connection between oldpeak and slope, identified during correlation analysis in EDA, and was validated by clinical literature (Gibbons et al 1997, 2002). ST depression magnitude and morphology are clinically interdependent based on ACC/AHA Exercise Testing Guidelines. This supported the creation of the st_severity feature, and replaced two correlated features (oldpeak and slope) thus resolving the zero inflation issue and multicollinearity simultaneously.

PCA assessment confirmed that dimensionality reduction was not needed in the continuous variables as the variance was evenly distributed and no clear elbow point was identified (see Figure 9).

SMOTE was explored as planned and will be compared with a set without SMOTE to determine whether model performance changes.

Variance Inflation Factor was applied to confirm and assess any remaining multicollinearity. No values exceeded 5.0, confirming that our variables are independent.

I believe these findings support the project's argument that model development for clinical support requires domain knowledge and clinical validation at every stage, not just the point of model evaluation.

Next Steps:

Next steps for analysis and modeling involve using the training sets created, X_train_encoded and X_train_smote, and applying them to several models. The comparison will evaluate whether SMOTE improves relevant performance metrics. The training set with better performance will be used for final modeling and validation. Models to be used are Logistic Regression as baseline, Random Forest, XGBoost, Logistic regression with CSL (cost-sensitive learning), and XGBoost with CSL. There is currently potential for 10 model variations.

Stratified k-fold will be applied to ensure robust performance estimates given the small sample size and mean / std will be reported across folds. Evaluation metrics will be accuracy, precision, recall, specificity, F1, ROC-AUC, as well as the sensitivity and specificity targets selected

previously to determine which model configuration best aligns with clinical risk thresholds.

Threshold analysis will determine what cutoff is clinically appropriate and DCA (Decision Curve Analysis) will be conducted in downstream model analysis.

In a clinical decision support context where false negatives carry more patient outcome risk than false positives, the question remains the same: which model's error profile is most clinically appropriate and not just the most accurate?

Reference:

Gibbons RJ, Balady GJ, Beasley JW, Bricker JT, Duvernoy WF, Froelicher VF, Mark DB, Marwick TH, McCallister BD, Thompson PD, Winters WL Jr, Yanowitz FG, Ritchie JL, Cheitlin MD, Eagle KA, Gardner TJ, Garson A Jr, Lewis RP, O'Rourke RA, Ryan TJ. ACC/AHA guidelines for exercise testing: executive summary. A report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines (Committee on Exercise Testing). *Circulation*. 1997 Jul 1;96(1):345-54. doi: 10.1161/01.cir.96.1.345. PMID: 9236456.

Gibbons RJ, Balady GJ, Bricker JT, Chaitman BR, Fletcher GF, Froelicher VF, Mark DB, McCallister BD, Mooss AN, O'Reilly MG, Winters WL Jr, Gibbons RJ, Antman EM, Alpert JS, Faxon DP, Fuster V, Gregoratos G, Hiratzka LF, Jacobs AK, Russell RO, Smith SC Jr; American College of Cardiology/American Heart Association Task Force on Practice Guidelines (Committee to Update the 1997 Exercise Testing Guidelines). ACC/AHA 2002 guideline update for exercise testing: summary article: a report of the American College of Cardiology/American Heart Association Task Force on Practice Guidelines (Committee to Update the 1997 Exercise Testing Guidelines). *Circulation*. 2002 Oct 1;106(14):1883-92. doi: 10.1161/01.cir.0000034670.06526.15. PMID: 12356646.

Appendix:

Figure 7: Q-Q Plots - Continuous Variables Normality Assessment

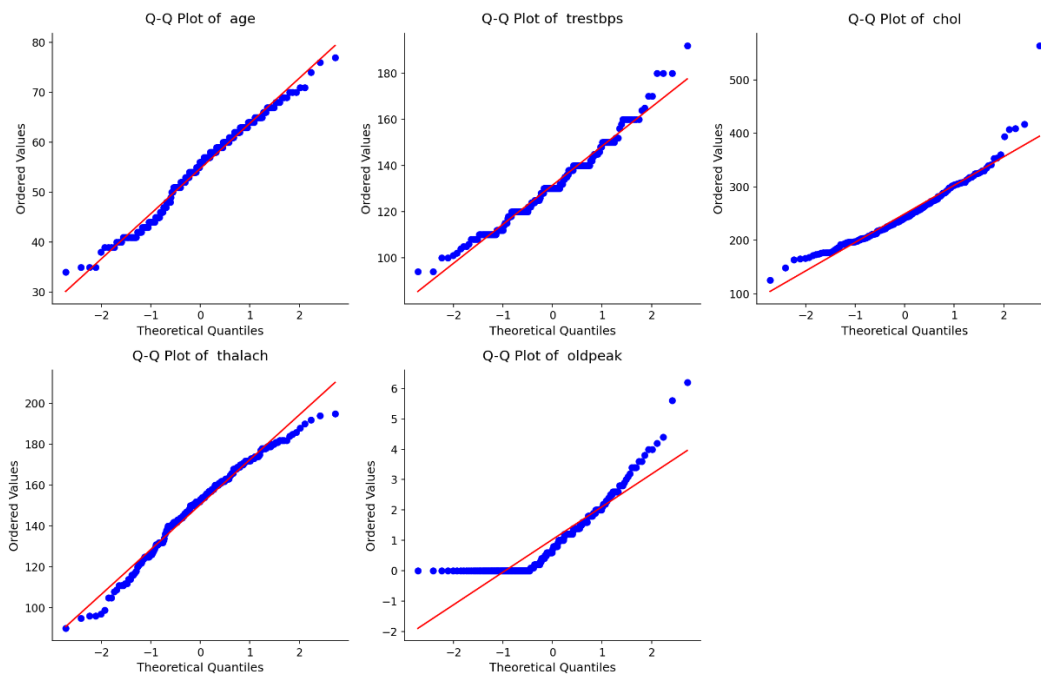


Figure 7: Q-Q plots for five continuous variables. Points on red diagonal indicate normality. Oldpeak shows departure from normal, consistent with right skew in EDA. Cholesterol shows an extreme outlier at upper tail. Age and maximum heart rate show normality despite failing p-value on Shapiro-Wilk test.

Figure 8: Q-Q Plots - Continuous Variables Normality Assessment Post Transformation

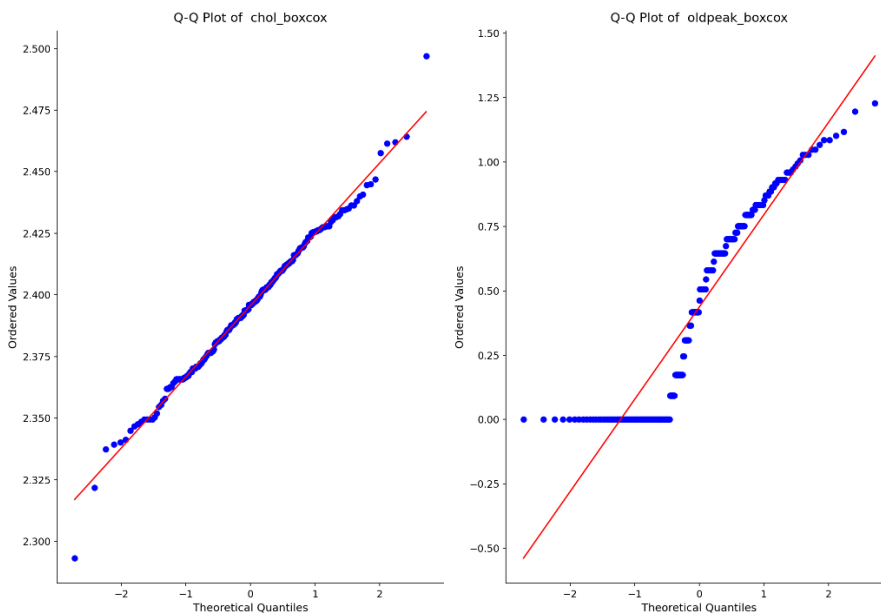


Figure 8: Q-Q plots for cholesterol and oldpeak after Box-Cox transformation. Cholesterol had improved p-value of 0.505 on Shapiro-Wilk. Oldpeak shows improvement but retains zero inflation pattern. Compare with Figure 7.

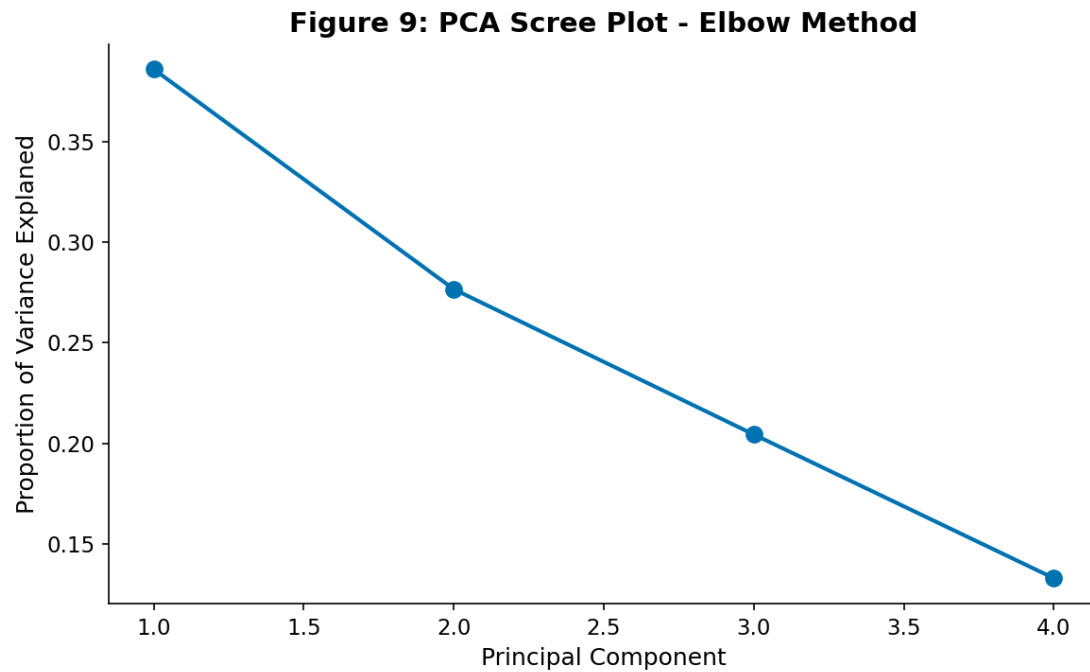


Figure 9: PCA scree plot using elbow method applied to all four continuous variables. Plot shows variance explained per component from PC1 through PC4, 38.6% $\rightarrow$ 13.3%. The absence of a clear elbow indicates variance is distributed across all four variables and confirms PCA will not reduce feature space.

Data Dictionary – Updated

Variable	Type	Description	Values/Range
age	Continuous	Age of patient in years	29 - 77, note: standardized via StandardScaler during preprocessing
sex	Categorical	Biological sex	0 = Female, 1 = Male
cp	Categorical	Chest pain type	1 = Typical Angina, 2 = Atypical Angina, 3 = Non-Anginal Pain, 4 = Asymptomatic
trestbps	Continuous	Resting blood pressure on admission (mmHg)	94 - 200, note : standardized via StandardScaler during preprocessing
chol_boxcox	Continuous	Box - Cox transformed cholesterol serum	standardized Original range: 126-564 mg/dl, transformed and standardized via StandardScaler
fb	Categorical	Fasting blood sugar > 120 mg/dl	0 = Normal, 1 = Elevated
restecg	Categorical	Resting electrocardiographic results	0 = Normal, 1 = ST-T Abnormality, 2 = LV Hypertrophy
thalach	Continuous	Maximum heart rate achieved during exercise (bpm)	71 - 202, note: standardized via StandardScaler during preprocessing
exang	Categorical	Exercise induced angina	0 = No, 1 = Yes
st_severity	Categorical	ST depression and slope of peak exercise severity index	0 = Normal, 1 = Equivocal, 2 = Positive, 3 = Strongly Positive
ca	Categorical	Number of major vessels colored by fluoroscopy	0 - 3
thal	Categorical	Thalassemia type	3 = Normal, 6 = Fixed Defect, 7 = Reversible Defect
target	Binary Target	Cardiovascular disease diagnosis	0 = No CVD, 1 = CVD Present

Project Details:

Language: Python, R with extensions, used in Google Colab.

Additional resources: scikit-learn, pandas, NumPy, matplotlib, seaborn, XGBoost.

GitHub Repository for this project can be found here:

https://github.com/mnooney/CVD_Capstone